INCH-POUND

MIL-B-23964D
AMENDMENT 5
23 NOVEMBER 1999

MILITARY SPECIFICATION

BOLT, SELF-RETAINING, POSITIVE LOCKING

This amendment forms a part of MIL-B-23964D, dated 10 august 1990, and is approved for use by all Departments and Agencies of the Department of Defense.

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Beneficial comments block: Delete "Naval Air Engineering Center, SESD (Code 53)" and substitute "Naval Air Warfare Center Aircraft Division Lakehurst, Systems Requirements Department, Code SR3."

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3.2 Delete and Substitute: "3.2 Qualification. The bolts furnished under this specification shall be produced using the same manufacturing methods as used to produce bolts that have passed qualification tests. The purchaser may apply any or all tests specified herein to any product represented as meeting this specification, at any time, to verify conformance."

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TABLE II: Delete locking balls push-out force column and substitute:

Locking Balls Push-out Force (lb. min)
10
20
20
20
40
60
60
80
80
80
80

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Delete paragraph 3.6.2 and substitute the following:

"3.6.2 Stress durability (class 3). Bolts shall withstand the specified load for the specified time without failing or developing cracks (see 4.5.22)."

Add new paragraph 3.6.10:

3.6.10 Stress corrosion (class 1 only). Bolts shall withstand the various stress corrosion conditions without cracking (see 4.5.24).

4.3.1 Delete and Substitute: "4.3.1 Qualification samples. Test samples shall be of the same material and manufactured by the same method as production parts. The number of bolts to be subjected to qualification tests shall be sufficient to perform the examinations identified in 4.3. Bolts shall have a grip length of 4D. Nuts in accordance with MS21133 or equivalent and washers (plain) in accordance with MS20002 shall be used for testing sizes 0.1900 through 0.5000 inches. Nuts in accordance with MS21133 (except non-self-locking) or equivalent in strength, and washers (plain) in accordance with MS20002 shall be used for testing sizes 0.5625 inches and larger."

"TABLE V: Change "Stress durability (except Class 4)" to "Stress durability (Class 3)"

ADD: "Stress Corrosion (Class 1)"

4.3.2 Delete and Substitute: "4.3.2.2 Certified Test Report. The manufacturer shall maintain on file a certified test report showing that the manufacturer's product satisfactorily conforms to this specification. The test report shall include, as a minimum, actual results of the tests specified herein. The manufacturer shall also maintain a dated drawing that completely describes the manufacturer's product. The drawing shall specify all dimensions and tolerances, material composition, coatings or platings applied, and heat treatment. The drawing shall also specify the manufacturer's part number for each size. Requalification will be required in the event any change is made in the product design, construction, materials, method of manufacture, heat treatment, finish, lubricant, or manufacturer's part number or designation."

"TABLE VI: Change "Stress durability (except Class 4)" to "Stress durability (Class 3)"

"TABLE VI: Delete "100%" from the Sampling plan column, for Locking ball push-out force, and substitute "R".

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4.4.2.1: Delete in its entirety and substitute:

"A production lot shall consist of finished bolts that are of the same head configuration, shank diameter, grip length, thread size, material, heat treatment by the same method in one continuous run and submitted to inspection at the same time."

4.5.3 Delete in its entirety and substitute:

"Each bolt shall be 100 percent functionally checked from both the unlocked and locked positions to determine if the operation is correct, without the loss of any element. Each bolt shall be 100 percent inspected for presence and containment of the locking elements and retention devices."

4.5.5: Add as last sentences:

"The sample shall be 10 percent of the lot size, minimum of 5 bolts with a maximum of 15 bolts. There shall be no failures.

Add new paragraph 4.5.24:

"4.5.24 Stress corrosion. The stress corrosion test shall be conducted on a minimum of one sample per lot using the procedure called out in accordance with MIL-TD-1312, Test 9. The induced load shall be 75 percent of the tested ultimate tensile strength of the bolts. Free running, cadmium plated MS21133 or equivalent nuts shall be used with self-retaining bolts for this test. The bolts shall be in the stressed condition for 72 hours. The test shall be continued an additional day for each day that the examination was not performed. Nuts required for this test shall be new and shall not be used more thanonce in the performance of this test.

Add as new paragraph 6.2.2:

"6.2.2 Inventory. Inventory produced before 1 January 1991 may be used until stock is exhausted."

6.3 Delete and Substitute: "6.3 Qualification. Awards will be made only for products that meet the qualification requirements of this document at the time set for opening of bids. Bolts that have not been previously qualified via government-certified qualification tests and reports may be acquired, provided that certified qualification test reports for these bolts are retained on file by the manufacturer along with the quality conformance inspection data, and they are made available to the buyer upon request."

6.3.1 Delete paragraph.

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NOTE: The margins of this amendment are marked with vertical lines to indicate where changes (additions, modifications, and corrections) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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